



AASD4011: Mathematical Concepts for Deep Learning

Gradient Descent Deep Dive

Definitions, Algorithms, and Examples

Derivative of a Function

Definition (limit form): For a function $f(x)$, its derivative at x is:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

- **Interpretation:**

- Geometrically, $f'(x)$ is the slope of the tangent line to $y=f(x)$ or we call $f'(x)=m$.
- Physically, if x is time and $f(x)$ is position, then $f'(x)$ is velocity.

- **Example:**

$$f(x) = x^2 \implies f'(x) = 2x$$

Partial Derivative

- Definition: How f changes when varying one coordinate at a time:

$$\frac{\partial f}{\partial x_i}(x_1, \dots, x_n) = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, \dots, x_i + \Delta x_i, \dots, x_n) - f(x_1, \dots, x_n)}{\Delta x_i}$$

- Intuition:
 - Hold all other x_j fixed, and see how changing x_i alone affects f .
- Example:

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + \cos(y)$$

3. Gradient of a Function

Definition: The gradient is the Vector of all partial derivatives, we call it partial derivatives.

- **Properties:**

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right]^\top$$

- Direction of steepest ascent: $\nabla f(\mathbf{x})$ points uphill.
- Magnitude: $\|\nabla f(\mathbf{x})\|$ tells how fast f increases.
- **Geometric picture:** In 2D, gradient is an arrow on the surface $z=f(x,y)$.

4. Gradient Descent Algorithm

- Gradient descent finds a local minimum by moving opposite to the gradient:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$$

- η is the learning rate (step size).
- Stop when $\|\nabla f(\mathbf{x}_t)\|$ is small or after fixed iterations.

Putting All Together

- Derivative $\rightarrow$ single slope; measures instantaneous rate of change.
- Partial derivative $\rightarrow$ rate of change for one variable, holding others fixed.
- Gradient $\rightarrow$ **full slope vector in multiple dimensions.**
- Gradient descent $\rightarrow$ iterative algorithm using the gradient to find minima or down hill of the function.

Quick Numerical Example

- Minimize $f(x,y) = x^2 + 2y^2$
- $\nabla f = [2x, 4y]$
- Learning Rate: 0.1
- Iteration:

$$(x, y) \leftarrow (x, y) - 0.1 [2x, 4y]$$